

Syed Jawad Ali

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Technical Skills

Programming: Python, JavaScript, TypeScript
Frameworks: Django, FastAPI, Next.js
Databases: PostgreSQL, MySQL, SQLite, pgvector
Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Agentic AI and Multi-Agent Systems, Hallucination Mitigation
Retrieval & Vector Systems: ChromaDB, PostgreSQL Vector Store (pgvector), Hybrid Retrieval (Keyword + Vector), Embeddings, Similarity Search
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Reinforcement Learning (DQN)
Systems & Tools: Docker, Git, Redis, REST APIs, Scalable System Design, Distributed AI Systems, Agile (Jira)

Research Interests

Adaptive Retrieval-Augmented Generation (RAG) and Knowledge-Intensive AI
Reinforcement Learning for Large Language Models and Agent Coordination
Multi-Agent AI Systems and Adaptive Decision-Making
Trustworthy AI, Hallucination Mitigation, and Explainable LLMs
Scalable AI Systems, Production Machine Learning, and Intelligent Information Retrieval

Publications

Manuscript Submitted

RAGENT: A Deep Q-Network-Driven Multi-Agent Framework for Adaptive Retrieval-Augmented Generation

Syed Jawad Ali, Sabina Akhar

Submitted to a Springer peer-reviewed venue

Presents a Deep Q-Network (DQN)-based multi-agent Retrieval-Augmented Generation (RAG) framework that investigates adaptive coordination among planning, retrieval, search, evaluation, and response generation through reinforcement learning. The framework employs a structured multi-objective reward function to optimize factual grounding, hallucination reduction, response quality, and computational efficiency. Experimental evaluation demonstrates improvements in retrieval effectiveness, response reliability, and adaptive decision-making compared with conventional static RAG pipelines.

Manuscript Submitted

DQN-RAG: Deep Reinforcement Learning-Based Adaptive Retrieval for Multi-Hop Question Answering

Syed Jawad Ali, Arif Ur Rahman

Submitted to the International Conference on Digital Technologies and Transformation (ICoDT2 2026), hosted by NUST SEecs, Islamabad, Pakistan

Introduces a reinforcement learning-driven multi-agent RAG framework in which a Deep Q-Network autonomously learns adaptive coordination policies for retrieval and reasoning in multi-hop question answering. The framework utilizes a six-dimensional state representation and a multi-objective reward function that jointly optimizes factual grounding, retrieval effectiveness, hallucination reduction, response quality, and token efficiency. Experimental evaluation on the HotpotQA benchmark demonstrates improvements in Exact Match (EM), F1 score, and computational efficiency compared with baseline approaches.

Professional Development Experience

Joyn Digital

June 2021 – Present

Backend Engineer (Python, Django, FastAPI, AI/LLMs)

Islamabad, Pakistan

- Designed and deployed scalable backend architectures for production-level AI and LLM applications using Django, FastAPI, PostgreSQL, and modern cloud deployment practices.
- Developed production-grade Retrieval-Augmented Generation (RAG) systems using hybrid retrieval, query rewriting, re-ranking, and confidence scoring to improve factual grounding, retrieval quality, and response reliability.
- Implemented reinforcement learning-based coordination strategies using Deep Q-Networks (DQN), enabling adaptive decisions for retrieval, web search, reasoning, and response generation in multi-agent RAG systems.

- Engineered multi-agent LLM architectures with specialized planning, retrieval, reasoning, search, evaluation, and orchestration components using LangGraph and modern agent-based frameworks.
- Optimized semantic retrieval pipelines using transformer embeddings, pgvector, FAISS, and PostgreSQL, improving retrieval precision, scalability, and contextual relevance for enterprise AI applications.

BUIC Research Lab

Jan 2019 – May 2021

Machine Learning Research Assistant

Islamabad, Pakistan

- Conducted research on machine learning algorithms using TensorFlow, Keras, and scikit-learn for classification, clustering, optimization, and predictive modeling tasks.
- Investigated natural language processing techniques using transformer-based models (BERT) and deep learning architectures for text representation, document classification, and performance evaluation.
- Performed data preprocessing, feature engineering, statistical analysis, and experimental evaluation using Python, Pandas, NumPy, and Matplotlib to support reproducible machine learning research.

NextIn Technologies

August 2016 – Dec 2018

Software Engineer (Backend Systems)

Islamabad, Pakistan

- Developed and maintained scalable backend systems using ASP.NET MVC, RESTful APIs, and relational databases with emphasis on software quality, performance, and maintainability.
- Collaborated with multidisciplinary teams to translate business requirements into robust software solutions, establishing a strong foundation in software engineering principles later applied to AI system development.

Projects

DQN-RAG: Reinforcement Learning-Based Multi-Agent Retrieval System | Django, LangGraph, n8n, PyTorch, Next.js

- Developed a novel DQN-based adaptive Retrieval-Augmented Generation (RAG) framework that learns optimal retrieval actions based on query complexity and response quality feedback.
- Designed a LangGraph multi-agent system comprising planning, retrieval, web search, evaluation, and answer-generation agents for complex question answering tasks.
- Investigated reinforcement learning for retrieval optimization and hallucination mitigation through reward-driven agent coordination and iterative decision-making.
- Implemented a scalable production architecture using ChromaDB, PostgreSQL, Celery, Redis, Docker, and workflow automation for real-world deployment.

RAGENT: Intelligent AI Chatbot with Adaptive Retrieval | Django, LangGraph, PostgreSQL, VectorDB, Next.js,

- Developed a full-stack AI chatbot capable of dynamically selecting between knowledge retrieval and direct response generation based on query context and confidence signals.
- Implemented hybrid information access through document-based RAG and external web search, enabling accurate responses to both enterprise and real-time queries.
- Built a secure authentication and user management system featuring JWT authentication, OTP verification, password recovery, and role-based access control.
- Designed scalable REST APIs and integrated asynchronous processing to support responsive, production-ready conversational AI applications.

Multi-Agent AI System with Decision Intelligence (RAG + RL) | Django, LangGraph, VectorDB, PostgreSQL, Next.js

- Proposed and developed a multi-agent LLM framework consisting of Planner, Retrieval, and Response agents for complex query decomposition and execution.
- Designed and experimentally evaluated an RL-based decision module (state-action-reward formulation) for optimizing agent coordination, balancing accuracy, latency, and cost.
- Investigated adaptive reasoning strategies to reduce hallucination and improve response reliability through dynamic decision-making policies.

Intelligent RAG SaaS Platform (AI Knowledge Assistant) | Django, LangGraph, VectorDB, PostgreSQL, Next.js

- Designed and implemented a production-level Retrieval-Augmented Generation (RAG) system for enterprise knowledge retrieval over heterogeneous documents (PDF, CSV, Docs).
- Developed hybrid retrieval pipelines (keyword + vector) using PostgreSQL (pgvector), improving retrieval accuracy and grounding responses with citation-based generation.
- Explored techniques such as re-ranking, query rewriting, and confidence scoring to enhance factual consistency and reduce hallucination in LLM outputs.

AI-Enhanced Human Resource Management System | Django, FastAPI, PostgreSQL, Next.js

- Developed an HR analytics platform integrating machine learning models for salary prediction and performance evaluation.
- Applied data-driven modeling techniques to analyze employee attributes and improve decision-making processes in HR management.

Education

Bahria University

Islamabad, Pakistan

Master of Science in Computer Science (CGPA: 3.40)

2020

- **Relevant Coursework:** Data Science, Machine Learning, Deep Learning, Artificial Intelligence
- **Thesis:** AI-based Selection of Web Services in Cloud using Fuzzy Logic Techniques
- **Focus:** Intelligent Decision Systems, Optimization under Uncertainty, Machine Learning, and Artificial Intelligence

Federal Urdu University of Science and Arts

Islamabad, Pakistan

Bachelor of Science in Computer Science (CGPA: 3.01)

2016

- **Relevant Coursework:** Data Structures and Algorithms, Programming Fundamentals (C++, Python), Artificial Intelligence, Web and Mobile Application Development
- **Final Year Project:** Kids Basic Learning Mobile Application